

Dear Hiring Manager,

I am a second-year student in Computer Science and Engineering at Delft University of Technology, writing to express my interest in a Software Engineering Internship at [Company Name].

My interest in programming, especially algorithms and data structures, first began with my competitive programming experience at the National Olympiad in Informatics, where I earned a Silver Medal. I subsequently developed an interest in low-level architecture through the Computer Organisation university course, for which I got a 9/10, which led me to start a custom NES Emulator project in C++. My work has included writing core logic for the 6502 CPU and the 2C02 PPU, and implementing essential assembly debugging for the emulator, an experience confirming both my enthusiasm for hardware emulation and my appreciation for working within tight architectural constraints.

My degree has deepened that foundation considerably. Through modules like Algorithms & Data Structures, Software Quality & Testing and Information & Data Management, I built a strong technical and theoretical foundation, maintaining an 8.04/10 average throughout my first year. Moreover, in the second quarter, I collaborated in a team of four to develop FoodPal, a distributed client-server recipe organizer application. Using Java and Spring Boot, I built the core recipe management features and a full-text search system that handles multi-term queries and filtering. I also added a live language switch using JavaFX internationalization, extending the app to five languages, a feature that took more edge-case handling than I expected going in. Working alongside three teammates on a growing codebase required coordination and planning, as the code had to be legible and accessible to everyone.

Outside of my core studies, my passion for strategic problem-solving and mentorship continues to shape my approach to collaboration. Building upon my previous leadership experience as President of my high school's Game Development and Chess clubs, I currently serve as a mentor for the university chess club. This role has sharpened my ability to communicate complex concepts clearly and support others in a highly analytical environment through lessons on various openings and strategic concepts.

I'd welcome the chance to bring this mix of hands-on engineering experience and academic grounding to the team at [Company Name]. Thank you for considering my application, and I look forward to hearing from you.

Yours sincerely,

Melvin Abibula